

1 Pre-Check ILP Formulation

Let T be the set of tasks and C the set of cores. For each task $t \in T$, let $E_t \subseteq C$ denote its eligible cores, d_t its execution time, and p_t its effective period. For each core $c \in C$, let s_c denote its WCET scaling factor.

The decision variable is defined as

$$y_{t,c} = \begin{cases} 1, & \text{if task } t \text{ is assigned to core } c, \\ 0, & \text{otherwise.} \end{cases}$$

The auxiliary variable $U_{\max}$ denotes the maximum utilization over all cores.

Model

$$\min \quad U_{\max} \tag{1}$$

Subject to

Task assignment. Each task must be assigned to exactly one eligible core:

$$\sum_{c \in E_t} y_{t,c} = 1 \quad \forall t \in T. \tag{2}$$

Assignments to ineligible cores are forbidden:

$$y_{t,c} = 0 \quad \forall t \in T, c \in C \setminus E_t. \tag{3}$$

Core utilization bound. For each core, the total scaled utilization of the tasks assigned to that core must not exceed $U_{\max}$:

$$\sum_{t \in T: c \in E_t} \frac{d_t s_c}{p_t} y_{t,c} \leq U_{\max} \quad \forall c \in C. \tag{4}$$

Domains

$$y_{t,c} \in \{0, 1\} \quad \forall t \in T, c \in C, \tag{5}$$

$$U_{\max} \geq 0. \tag{6}$$